

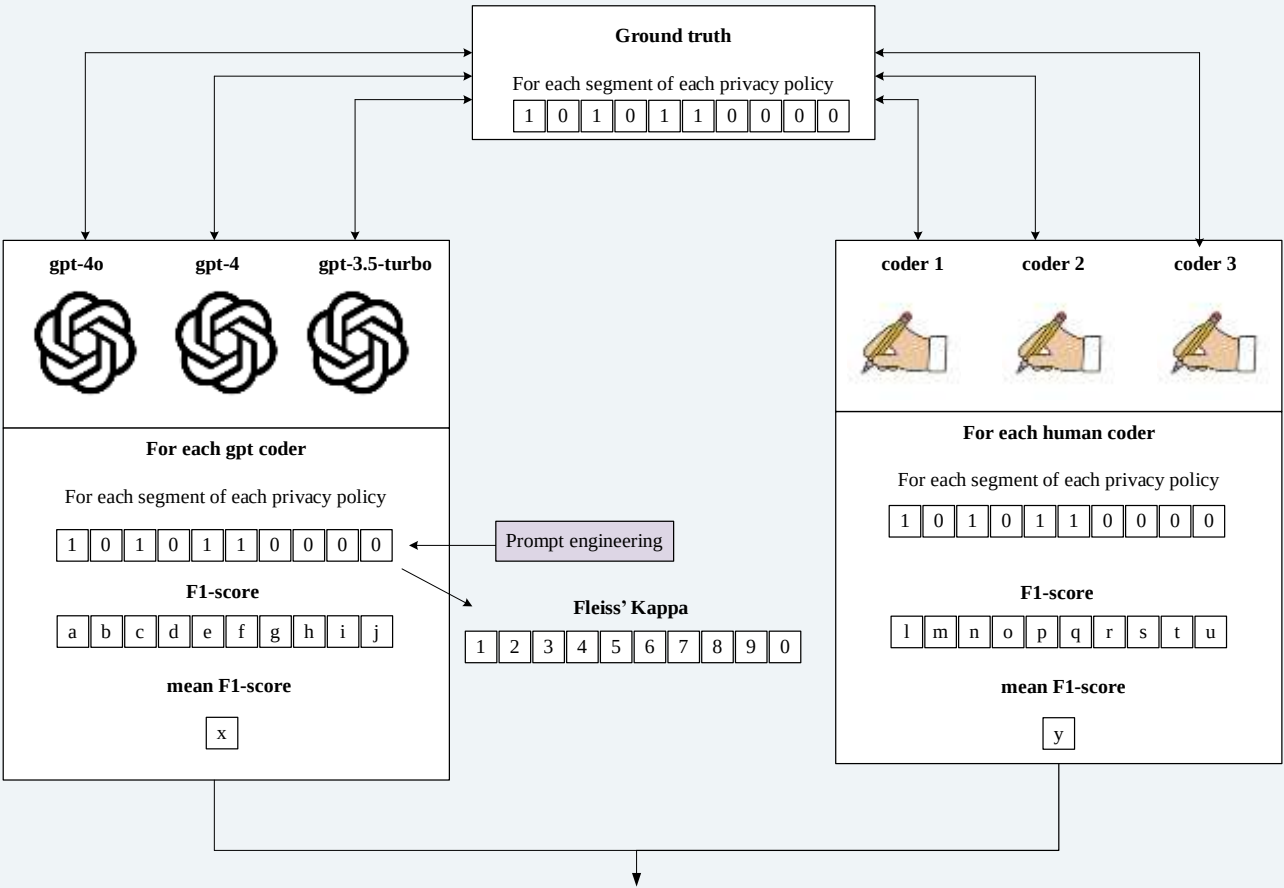
A close-up photograph of a hand holding a black fountain pen with gold accents, poised to write on a document. The document contains faint, printed text and a grid-like structure. The background is blurred, showing a desk and a computer monitor.

**ENHANCING PRIVACY POLICIES ANNOTATION:
BEYOND THE HUMAN THRESHOLD**

Escuela Politécnica Nacional – Universidad Politécnica de Madrid

August, 2024

Proposal 1: 3 gpts vs 3 humans



	coder 1	coder 2	coder 3
gpt-4o	p-value	p-value	p-value
gpt-4	p-value	p-value	p-value
gpt-3.5-turbo	p-value	p-value	p-value

Log_probs and confidence level ☀️

- Without confidence level
- Confidence level = 0.8
- Confidence level = 0.85
- Confidence level = 0.9
- Confidence level = 0.95

- Key differentiators:**
- API based LLM
 - Privacy policies focus
 - Multiclass and multilabel problem
 - Inter-coder agreement among LLMs
 - Zero shot capacity analysis
 - Strong statistical analysis
 - Techniques: Answer/question, One VS rest

Agreements

Corpus

- OPP-115: This corpus is suitable to carry out the research since it has available a codebook and individual coders' annotations.

Ground truth

- The csv records that starts with the letter 'C' from the 'consolidation' folder considering the three thresholds will serve as ground truth.

Openai API

- The research is based on Large Language Models (LLMs) using the openai API since these models have demonstrated high performance for Natural Language Processing (NLP).
- Each API request is independent from the previous requests.

Agreements

Prompt engineering

- We Will use the prompt structure suggested in previous study from UPM as baseline.

Log_probs

- Analyzing log_probs of resulting tokens is an important differentiator from previous studies.

To be defined

Number of gpts

- Are really necessary three gpts?
- Do we need to compare the coders' inter-agreement?
- Which gpt to use?

Segment evaluation vs whole policy evaluation

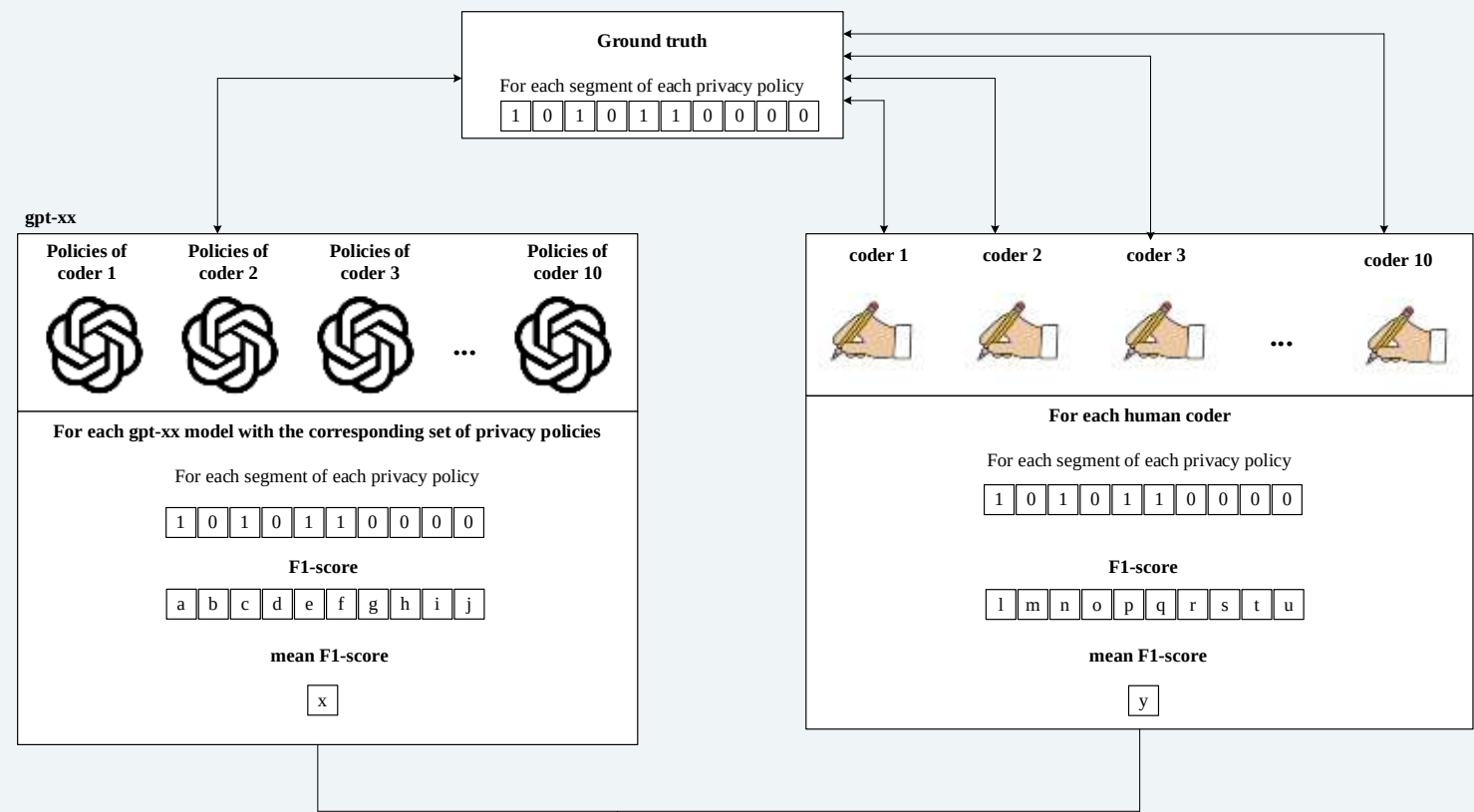
- Do we need to carry out an evaluation per segment or evaluate the whole policy text?
- Is the whole context really relevant for our problem?

Zero shot vs few shot

- Do we need to provide examples at each request?
- Is few shot required since we purpose to demonstrate LLM capacity to annotate privacy policites, that is, to create the corpus?

Proposal 2: 1 gpt vs 10 humans

ENHANCING PRIVACY POLICIES ANNOTATION: BEYOND THE HUMAN THRESHOLD



	coder 1	coder 2	coder 3	...	coder 10
gpt-4o	p-value	p-value	p-value	...	p-value

Log_probs and confidence level ☀

- Without confidence level
- Confidence level = 0.8
- Confidence level = 0.85
- Confidence level = 0.9
- Confidence level = 0.95

Key differentiators:

- API based LLM
- Privacy policies focus
- Multiclass and multilabel problem
- Zero shot capacity analysis
- Strong statistical analysis
- Techniques: Answer/question, One VS rest

Agreements

Zero shot

- We will use a zero shot approach due to the corpus does not expose examples and the research purposes to analyze the capacity of a gpt in comparison to the human performance.

A single gpt model

- A single gpt model will be used in order to carry out the research.

11th coder

- A gpt model that annotates all the privacy policies at segments-level will work as 11th coder (experiment 1).

12th coder

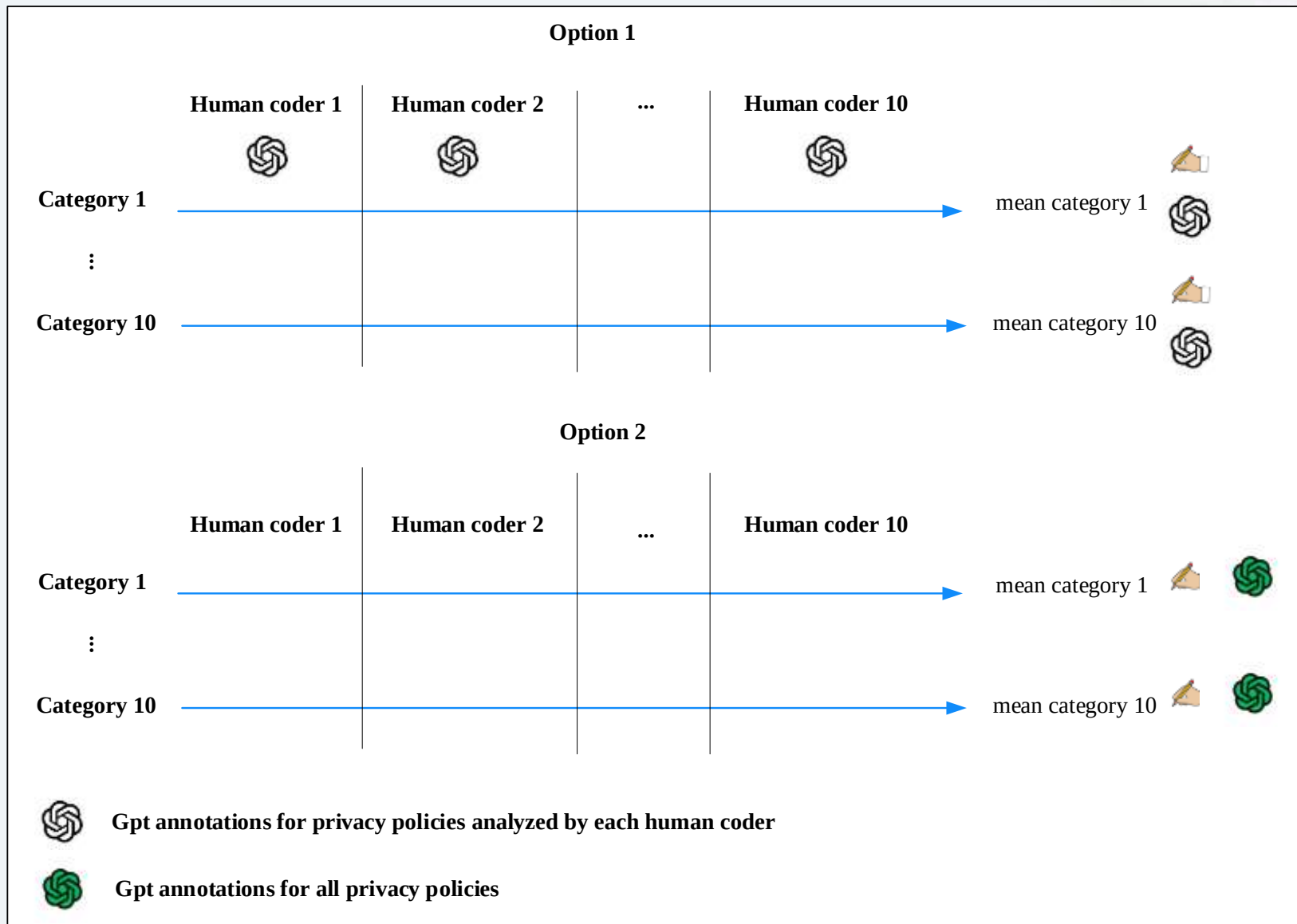
- A gpt model that annotates all the privacy policies at whole text-level will work as 12th coder (experiment 2).

To be defined

Evaluation of results

- For human performance we could take the mean of each category among the ten coders.
- For gpt performance we could calculate the metrics reached for each privacy policies set annotated by each human coder, and therefore calculate the mean.
- A general comparison could be made between the two aforementioned calculations.
- Another option can be to take the overall performance of the gpt for the whole privacy policies and compare it to the mean achieved by the coders.
- The selected method will be applied at segments-level annotations and whole text-level privacy policies annotations.

To be defined



Agreements

First proposal

- We will use the first option in order to evaluate the performance of human coders against gpt's performance.

Human annotators (coders)



Coder 1
(ID 82)



Coder 4
(ID 95)



Coder 5
(ID 103)



Coder 8
(ID 118)



Coder 2
(ID 84)



Coder 9
(ID 121)



Coder 3
(ID 88)



Coder 6
(ID 116)



Coder 7
(ID 117)



Coder 10
(ID 123)



Human coders' performance

Segment-level annotations

Coder 1 (ID 82)

F1-score category First Party Collection/Use: 0.31063829787234043
F1-score category Third Party Sharing/Collection: 0.21354166666666666
F1-score category User Choice/Control: 0.2923076923076923
F1-score category User Access, Edit and Deletion: 0.5507246376811594
F1-score category Data Retention: 0.3684210526315789
F1-score category Data Security: 0.8118811881188119
F1-score category Policy Change: 0.7941176470588236
F1-score category International and Specific Audiences: 0.8977272727272727
F1-score category Do Not Track: 0.9
F1-score category Other: 0.7751937984496123
Mean f1-score: 0.5914553253513958

Coder 3 (ID 88)

F1-score category First Party Collection/Use: 0.3397129186602871
F1-score category Third Party Sharing/Collection: 0.2165242165242165
F1-score category User Choice/Control: 0.31481481481481477
F1-score category User Access, Edit and Deletion: 0.4810126582278481
F1-score category Data Retention: 0.42424242424242425
F1-score category Data Security: 0.84
F1-score category Policy Change: 0.6913580246913581
F1-score category International and Specific Audiences: 0.8901098901098902
F1-score category Do Not Track: 0.7999999999999999
F1-score category Other: 0.6532438478747203
Mean f1-score: 0.5651018795145559

Coder 2 (ID 84)

F1-score category First Party Collection/Use: 0.34912718204488774
F1-score category Third Party Sharing/Collection: 0.2254335260115607
F1-score category User Choice/Control: 0.27906976744186046
F1-score category User Access, Edit and Deletion: 0.5230769230769231
F1-score category Data Retention: 0.4
F1-score category Data Security: 0.8247422680412373
F1-score category Policy Change: 0.6250000000000001
F1-score category International and Specific Audiences: 0.8457142857142858
F1-score category Do Not Track: 0.8571428571428571
F1-score category Other: 0.8111587982832618
Mean f1-score: 0.5740465607756874

Coder 4 (ID 95)

F1-score category First Party Collection/Use: 0.5929487179487178
F1-score category Third Party Sharing/Collection: 0.5454545454545454
F1-score category User Choice/Control: 0.34224598930481287
F1-score category User Access, Edit and Deletion: 0.25
F1-score category Data Retention: 0.5161290322580645
F1-score category Data Security: 0.8227848101265823
F1-score category Policy Change: 0.6666666666666666
F1-score category International and Specific Audiences: 0.97165991902834
F1-score category Do Not Track: 0.9473684210526316
F1-score category Other: 0.7906976744186047
Mean f1-score: 0.6445955776258966

Human coders' performance

Segment-level annotations

Coder 5 (ID 103)

F1-score category First Party Collection/Use: 0.5108225108225107
F1-score category Third Party Sharing/Collection: 0.5301204819277108
F1-score category User Choice/Control: 0.32460732984293195
F1-score category User Access, Edit and Deletion: 0.19999999999999998
F1-score category Data Retention: 0.34782608695652173
F1-score category Data Security: 0.6431718061674009
F1-score category Policy Change: 0.5915492957746478
F1-score category International and Specific Audiences: 0.9421487603305785
F1-score category Do Not Track: 0.9
F1-score category Other: 0.7370441458733206
Mean f1-score: 0.5727290417695623

Coder 6 (ID 116)

F1-score category First Party Collection/Use: 0.6061946902654868
F1-score category Third Party Sharing/Collection: 0.6027397260273973
F1-score category User Choice/Control: 0.3529411764705882
F1-score category User Access, Edit and Deletion: 0.4266666666666667
F1-score category Data Retention: 0.4285714285714286
F1-score category Data Security: 0.6355140186915887
F1-score category Policy Change: 0.3888888888888889
F1-score category International and Specific Audiences: 0.7846153846153847
F1-score category Do Not Track: 1.0
F1-score category Other: 0.7206611570247934
Mean f1-score: 0.5946793137222223

Coder 7 (ID 117)

F1-score category First Party Collection/Use: 0.6451612903225806
F1-score category Third Party Sharing/Collection: 0.5206349206349207
F1-score category User Choice/Control: 0.4651162790697675
F1-score category User Access, Edit and Deletion: 0.6666666666666666
F1-score category Data Retention: 0.4444444444444444
F1-score category Data Security: 0.7894736842105263
F1-score category Policy Change: 0.6857142857142856
F1-score category International and Specific Audiences: 0.9183673469387755
F1-score category Do Not Track: 0.6666666666666666
F1-score category Other: 0.7897727272727273
Mean f1-score: 0.6592018311941362

Coder 8 (ID 118)

F1-score category First Party Collection/Use: 0.6197530864197531
F1-score category Third Party Sharing/Collection: 0.5341812400635929
F1-score category User Choice/Control: 0.3393501805054152
F1-score category User Access, Edit and Deletion: 0.2784810126582279
F1-score category Data Retention: 0.3448275862068965
F1-score category Data Security: 0.6514285714285715
F1-score category Policy Change: 0.6075949367088608
F1-score category International and Specific Audiences: 0.9068825910931174
F1-score category Do Not Track: 0.923076923076923
F1-score category Other: 0.6758147512864494
Mean f1-score: 0.5881390879447809

Human coders' performance

Segment-level annotations

Coder 9 (ID 121)

F1-score category First Party Collection/Use: 0.666666666666667
F1-score category Third Party Sharing/Collection: 0.6403940886699507
F1-score category User Choice/Control: 0.5113636363636364
F1-score category User Access, Edit and Deletion: 0.4225352112676056
F1-score category Data Retention: 0.339622641509434
F1-score category Data Security: 0.8545454545454545
F1-score category Policy Change: 0.6666666666666666
F1-score category International and Specific Audiences: 0.9433962264150944
F1-score category Do Not Track: 1.0
F1-score category Other: 0.8179551122194514
Mean f1-score: 0.6863145704323961

Coder 10 (ID 123)

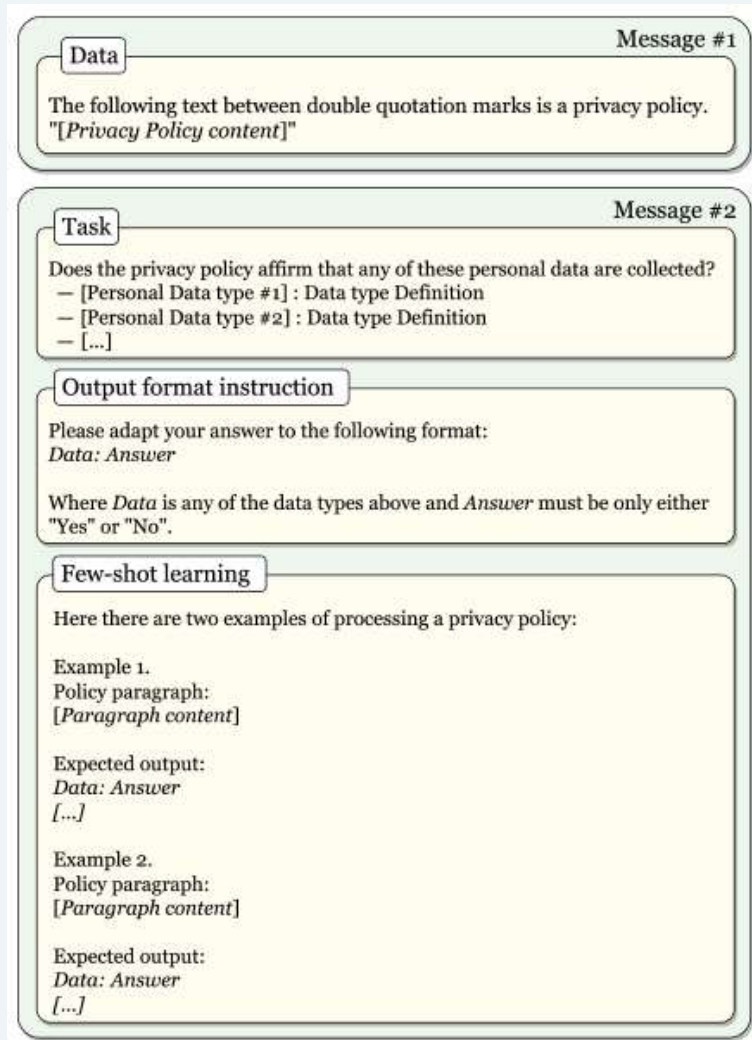
F1-score category First Party Collection/Use: 0.5512820512820513
F1-score category Third Party Sharing/Collection: 0.5346534653465346
F1-score category User Choice/Control: 0.326530612244898
F1-score category User Access, Edit and Deletion: 0.3571428571428571
F1-score category Data Retention: 1.0
F1-score category Data Security: 0.7692307692307693
F1-score category Policy Change: 0.4
F1-score category International and Specific Audiences: 0.823529411764706
F1-score category Do Not Track: 1.0
F1-score category Other: 0.763819095477387
Mean f1-score: 0.6526188262489202

JSON result structure

```
{
  "policy": 11,
  "categories": [
    {
      "category": 0,
      "log_prob": -3.6073914e-05
    },
    {
      "category": 1,
      "log_prob": -0.00015145572
    },
    {
      "category": 2,
      "log_prob": -0.0006130747
    },
    {
      "category": 3,
      "log_prob": -0.31534484
    },
    {
      "category": 4,
      "log_prob": -0.0014071364
    },
    {
      "category": 6,
      "log_prob": -0.6330793
    },
    {
      "category": 9,
      "log_prob": -0.02532874
    }
  ]
}
```

Prompt structure

Did you use the role system and role user for this prompt structure?



Role system

Role user

A close-up photograph of a hand holding a black pen with a gold nib, writing on a document. The document has a grid of small squares, possibly for data recording. The background is blurred, showing more of the document and some papers underneath.

Experiments Design

Segments-level Annotation

- Ground truth and human annotations at segments levels.
- One GPT request to OpenAI API for each segment.
- Calculus of the F1-score metric for each segment of the privacy policies annotated by each human.
- Graphical representation of the F1-score distributions achieved by each human and the GPT (with non-threshold and thresholds at 0.8, 0.85, 0.9, 0.95)
- Unidirectional Wilcoxon rank sum hypothesis test at a significance level of 0.05.

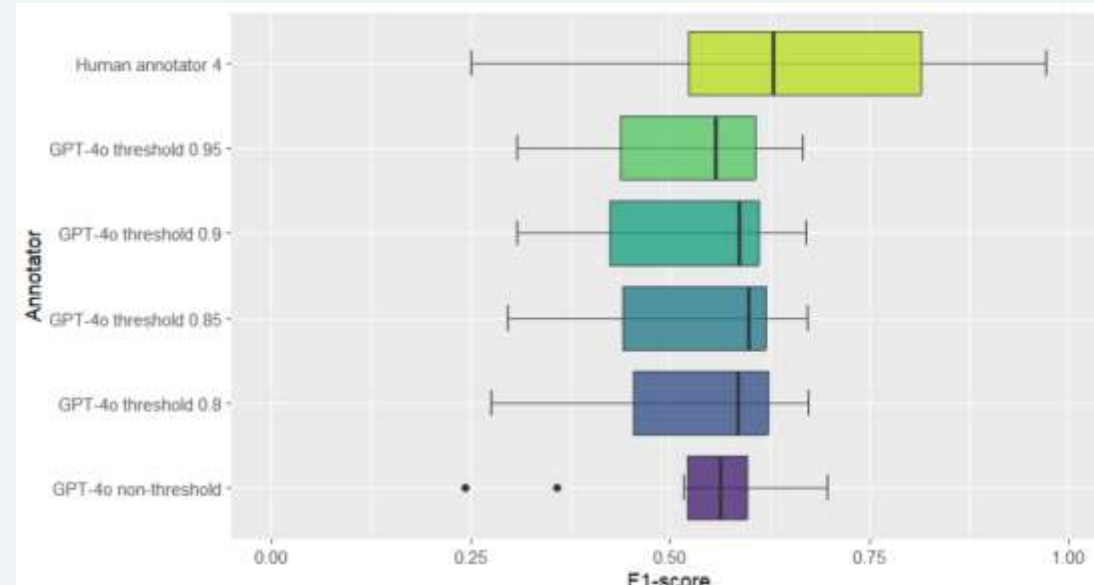
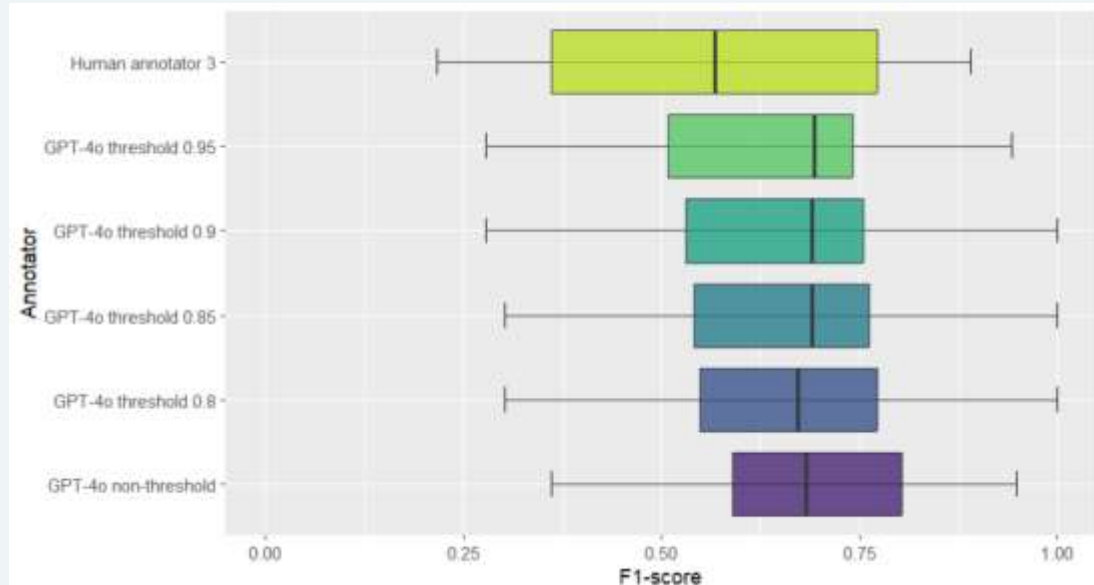
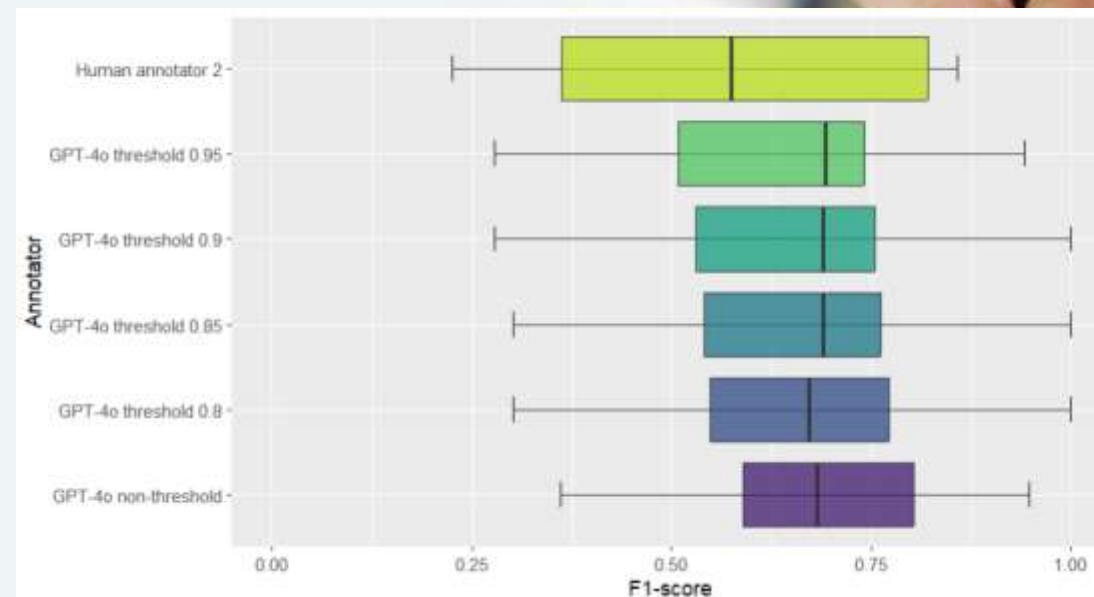
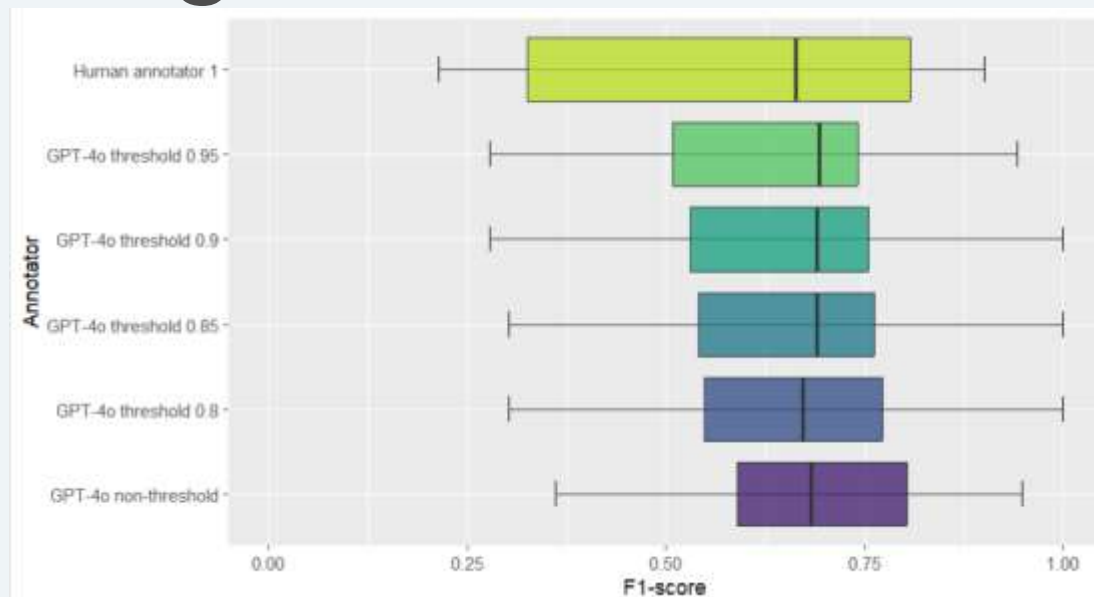
Whole-text level annotations

- Ground truth and human annotations adapted to whole-text level annotation (one appearance in a segment involves the annotation in the entire privacy policy).
- One GPT request to OpenAI API for each privacy policy.
- Calculus of the F1-score metric for each privacy policy of each human annotator.
- Graphical representation of the F1-score distributions achieved by each human and the GPT (with non-threshold and thresholds at 0.8, 0.85, 0.9, 0.95)
- Unidirectional Wilcoxon rank sum hypothesis test at a significance level of 0.05.

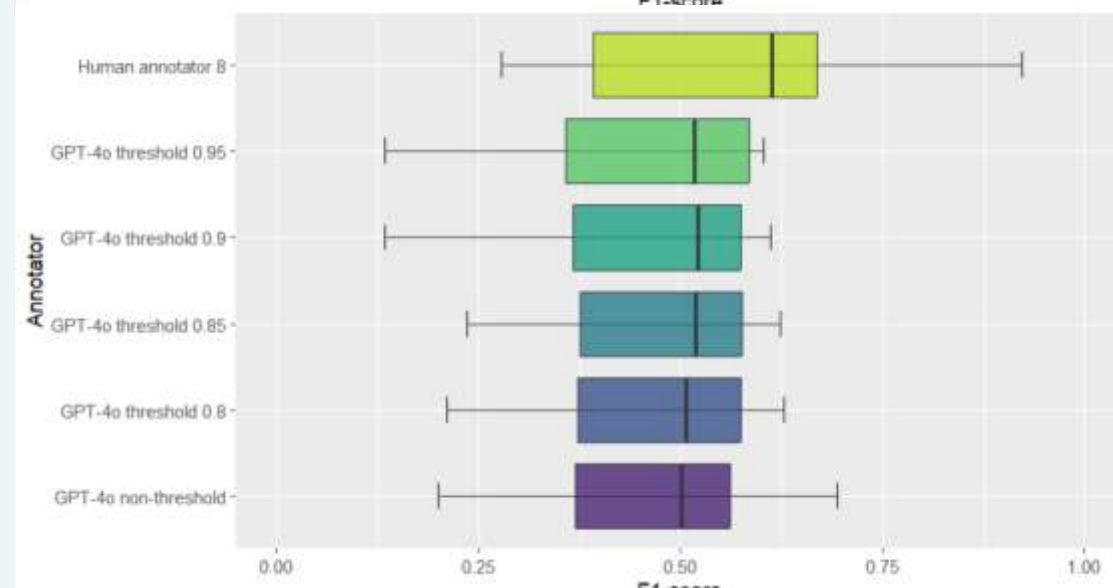
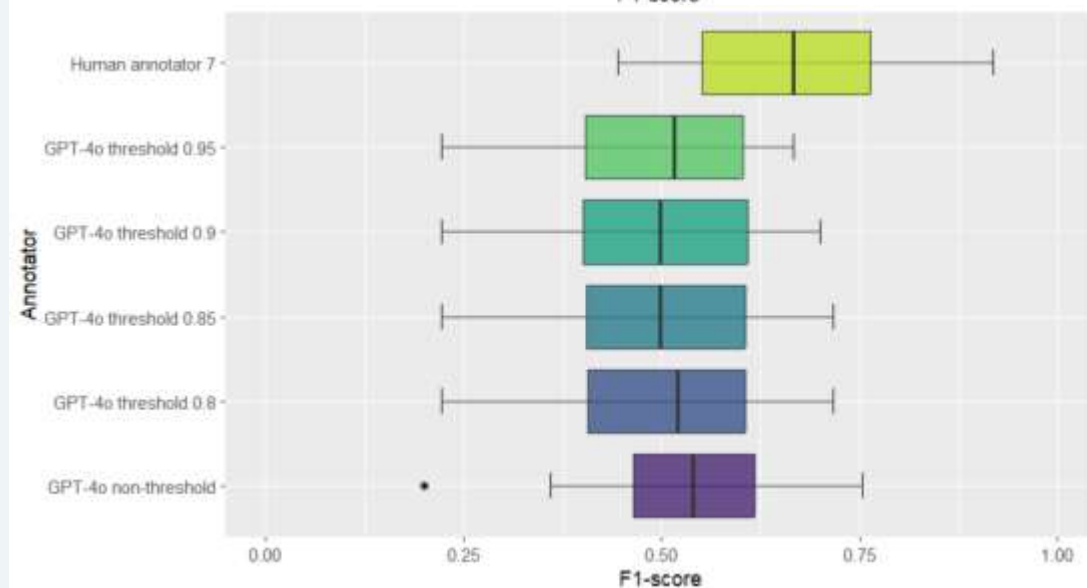
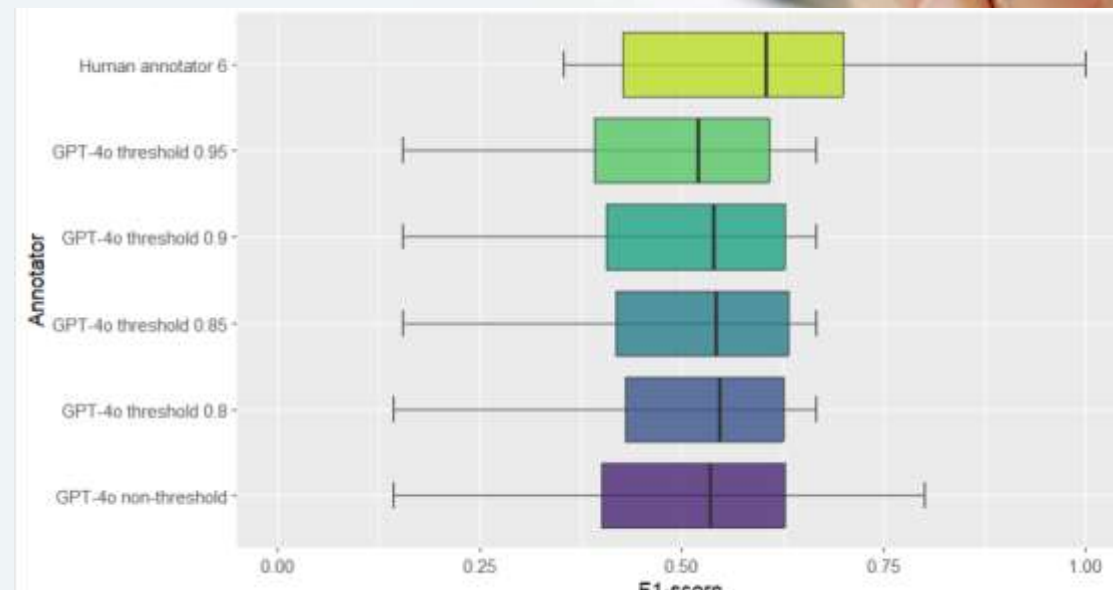
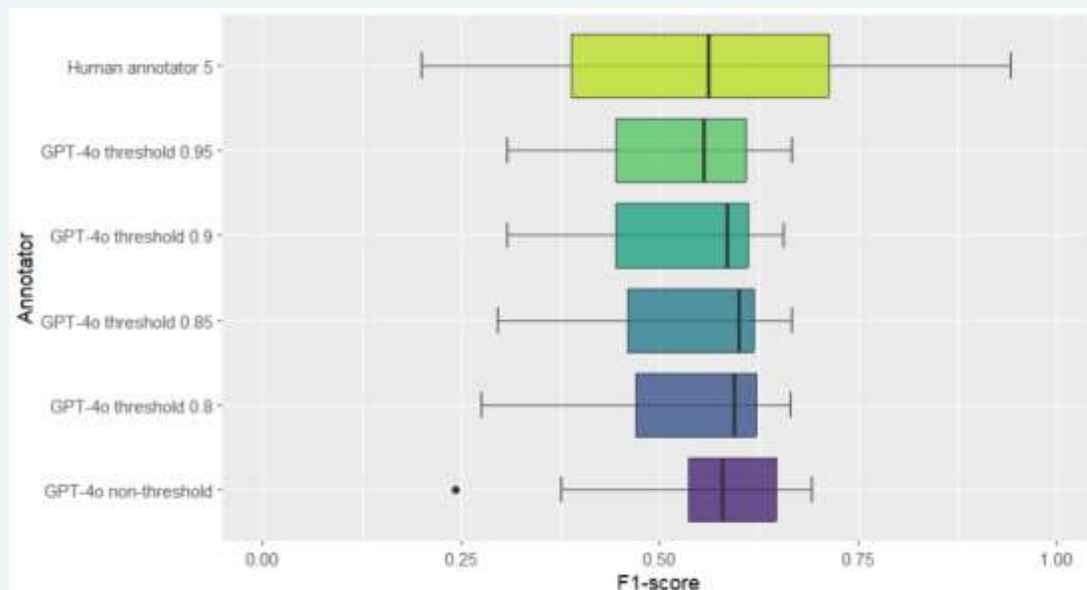
A close-up, shallow depth-of-field photograph of a person's hand holding a black fountain pen with a gold nib. The hand is positioned over a document with faint, illegible text. The background is a soft, out-of-focus white.

Results

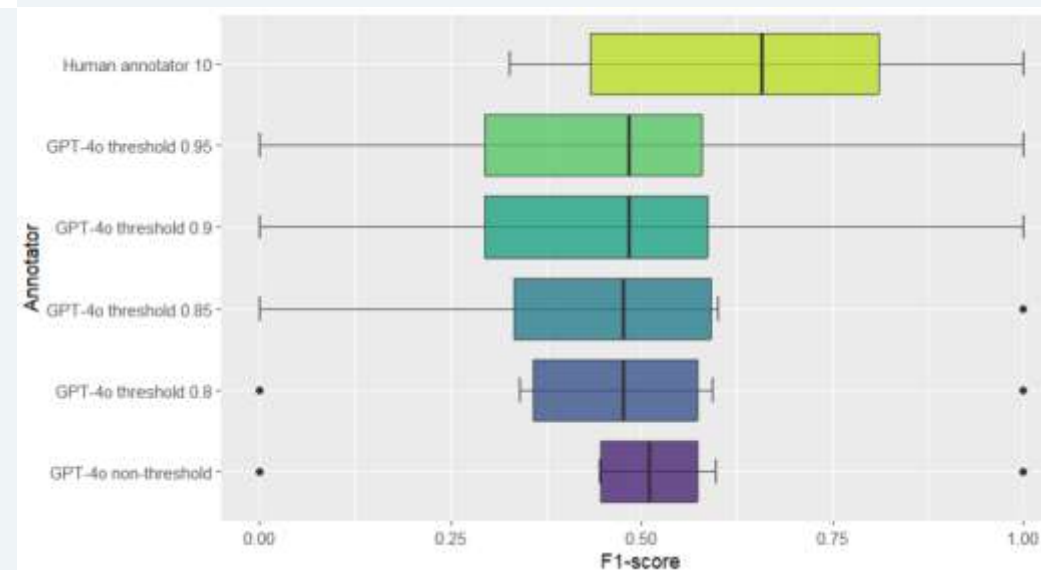
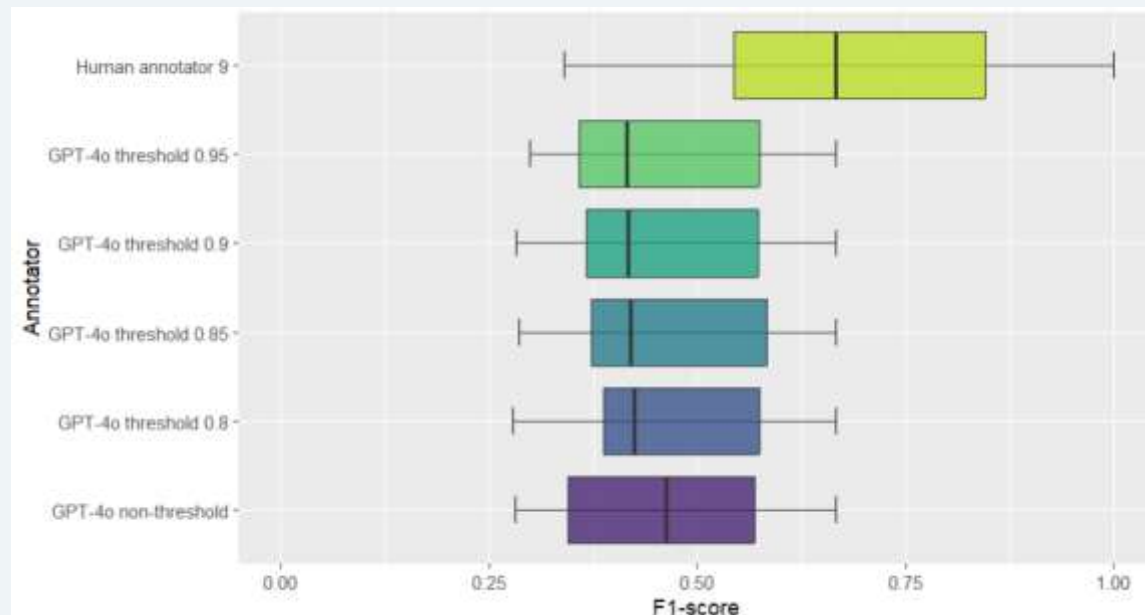
Segments-level Annotation



Segments-level Annotation



Segments-level Annotation

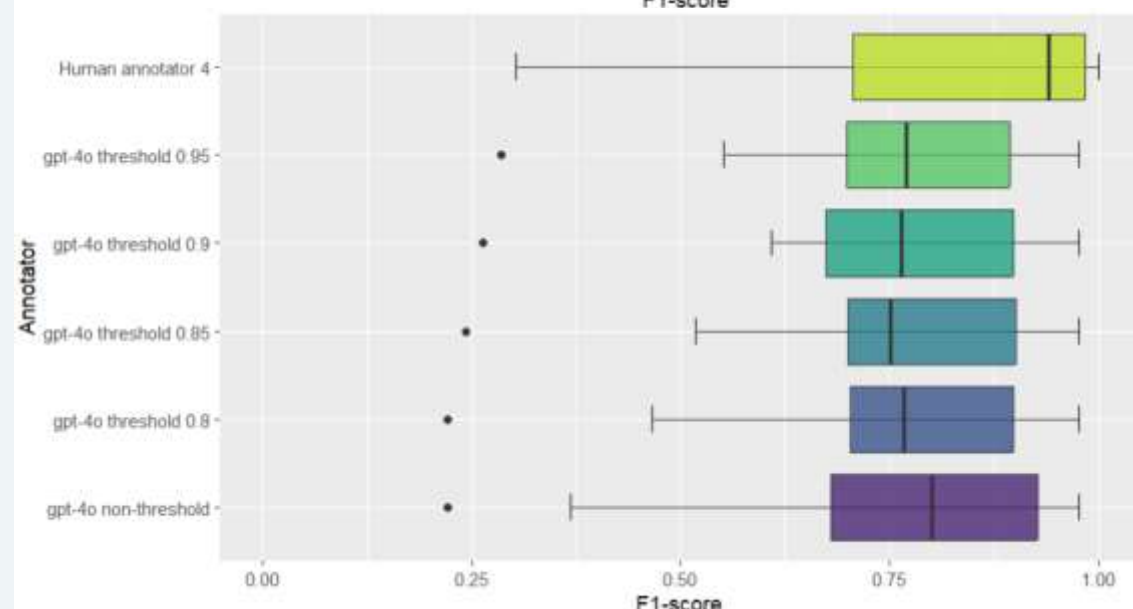
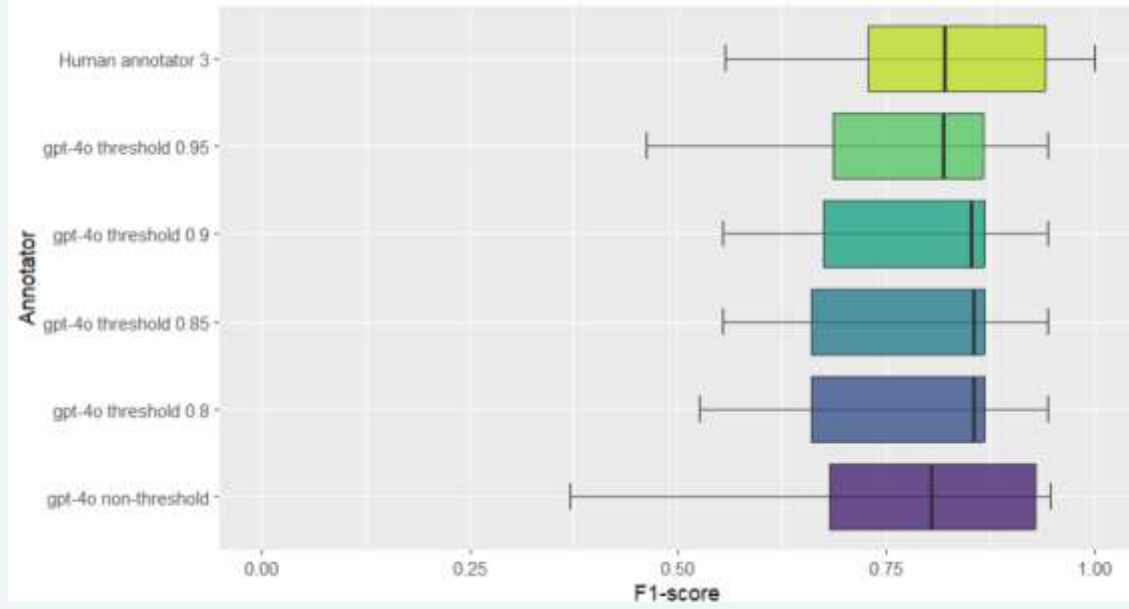
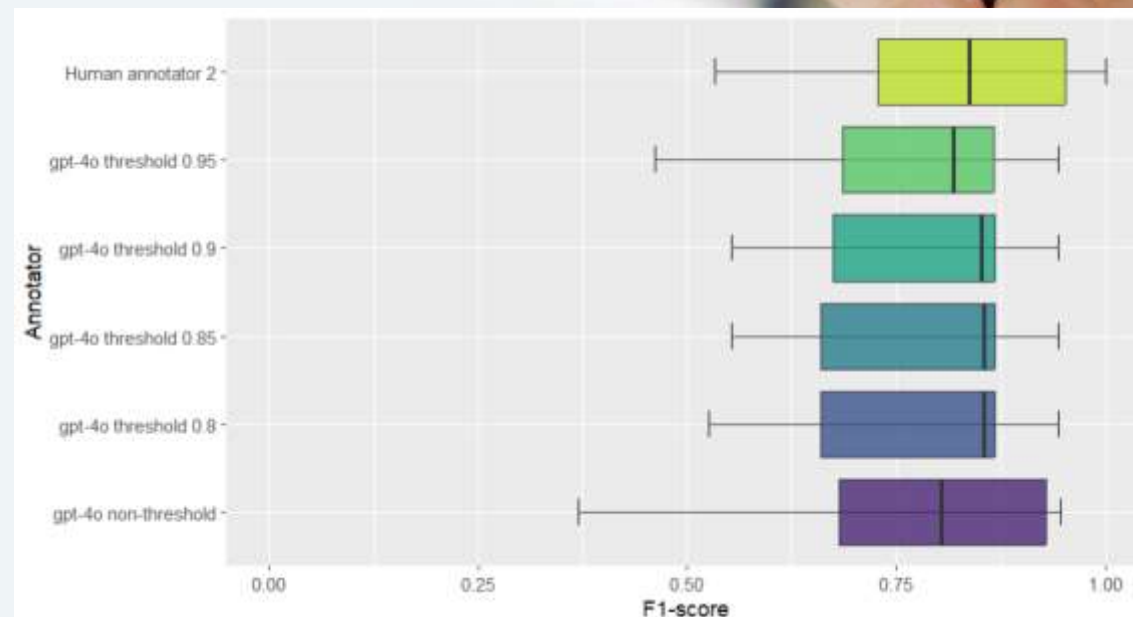
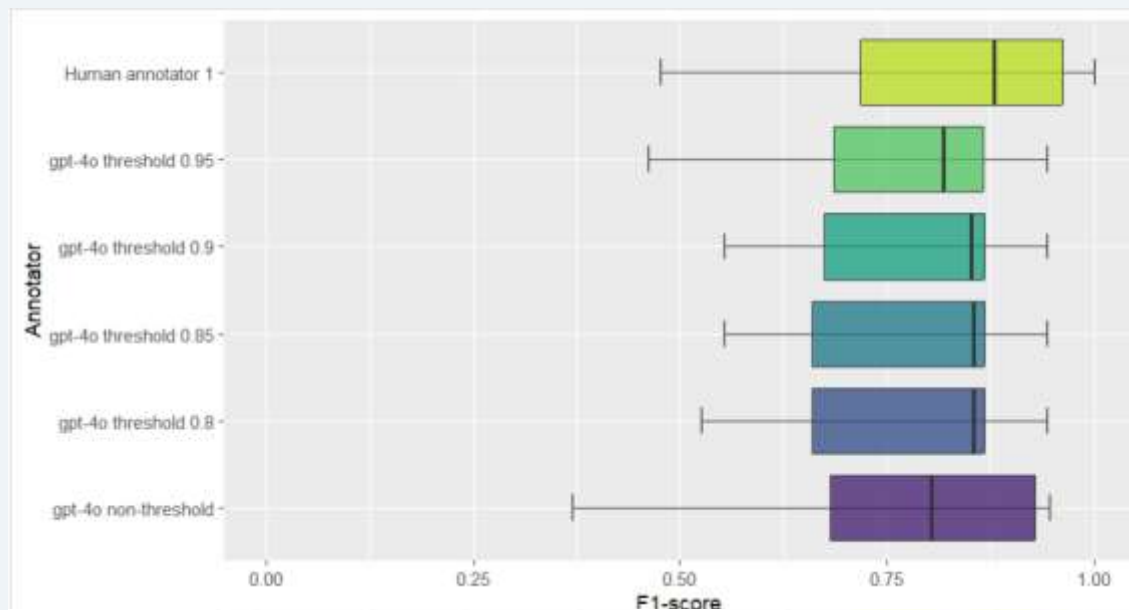


Segments-level Annotation

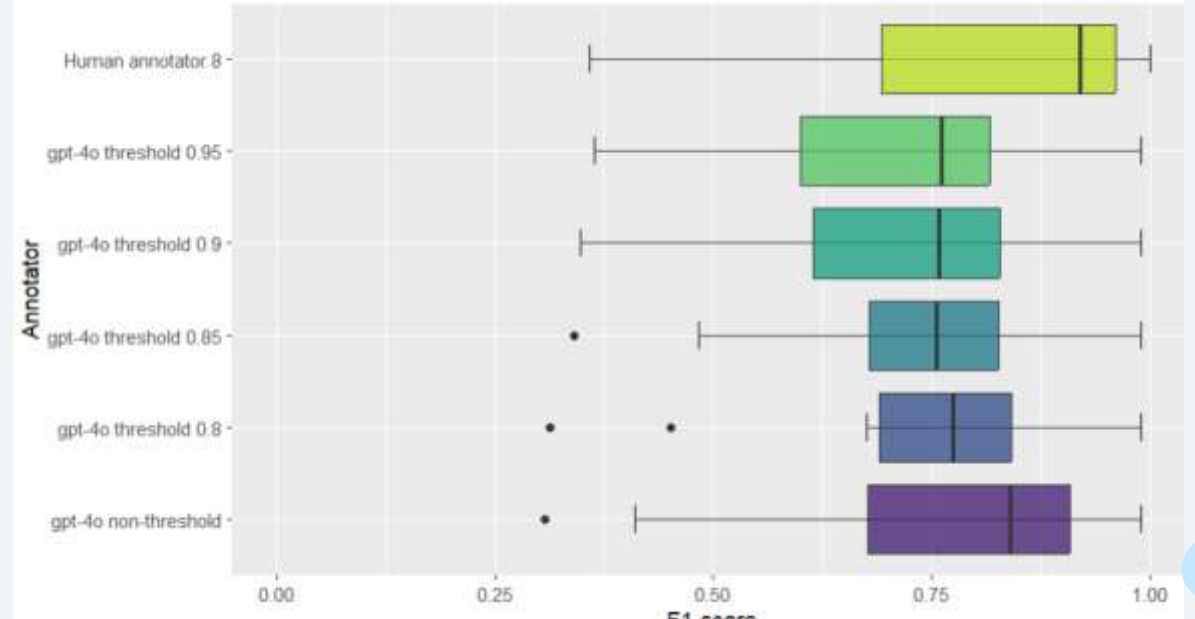
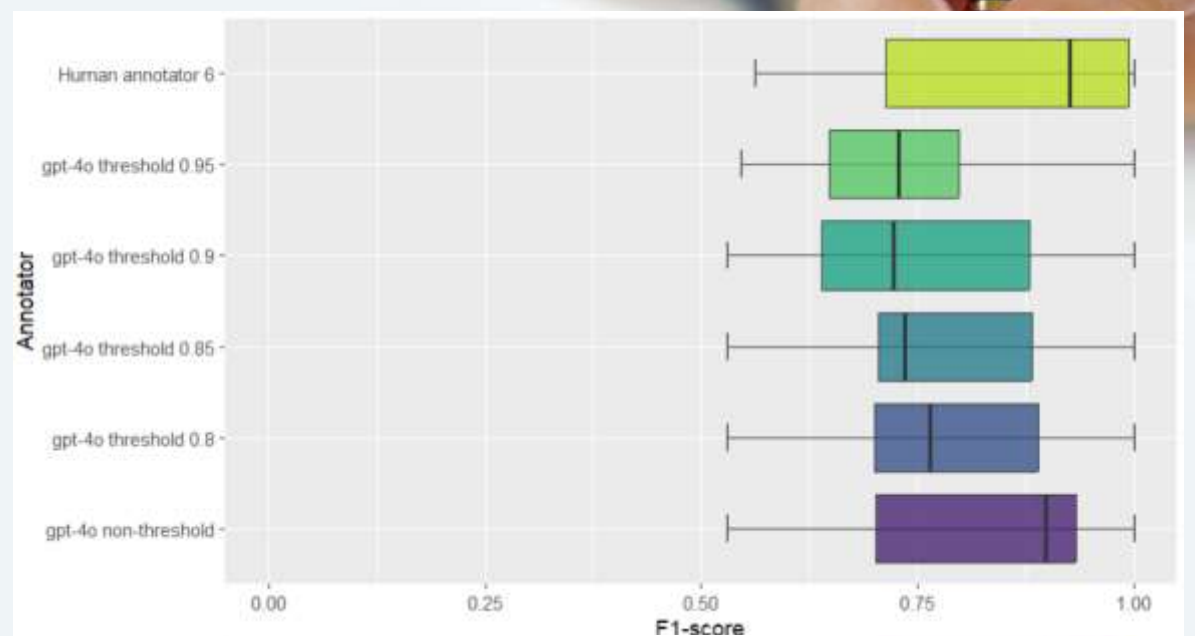
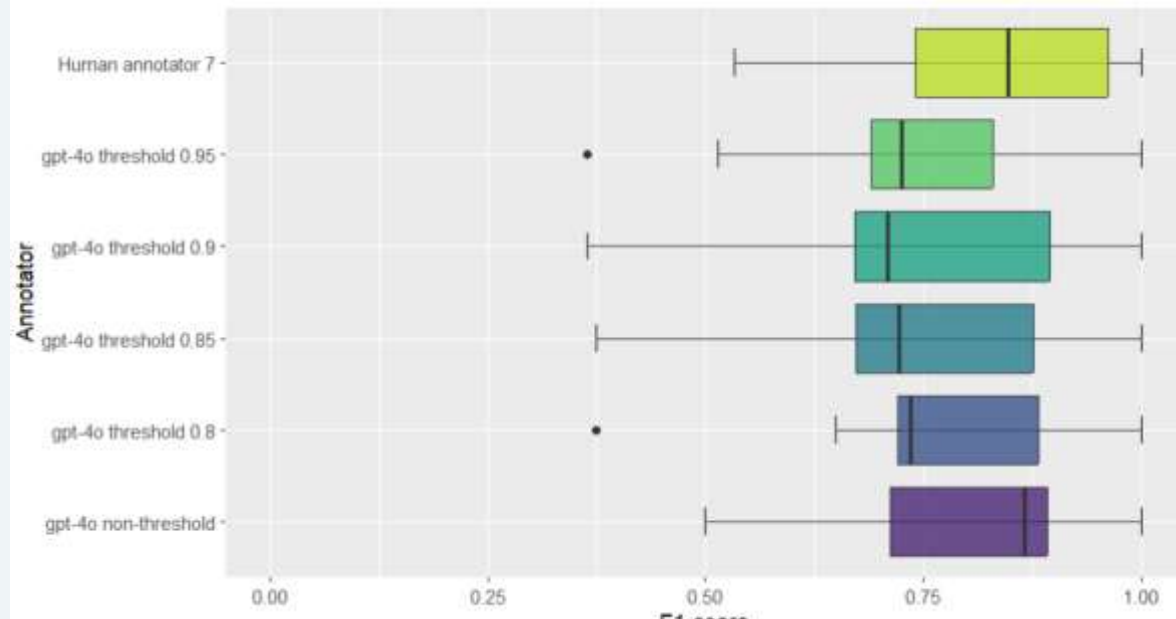
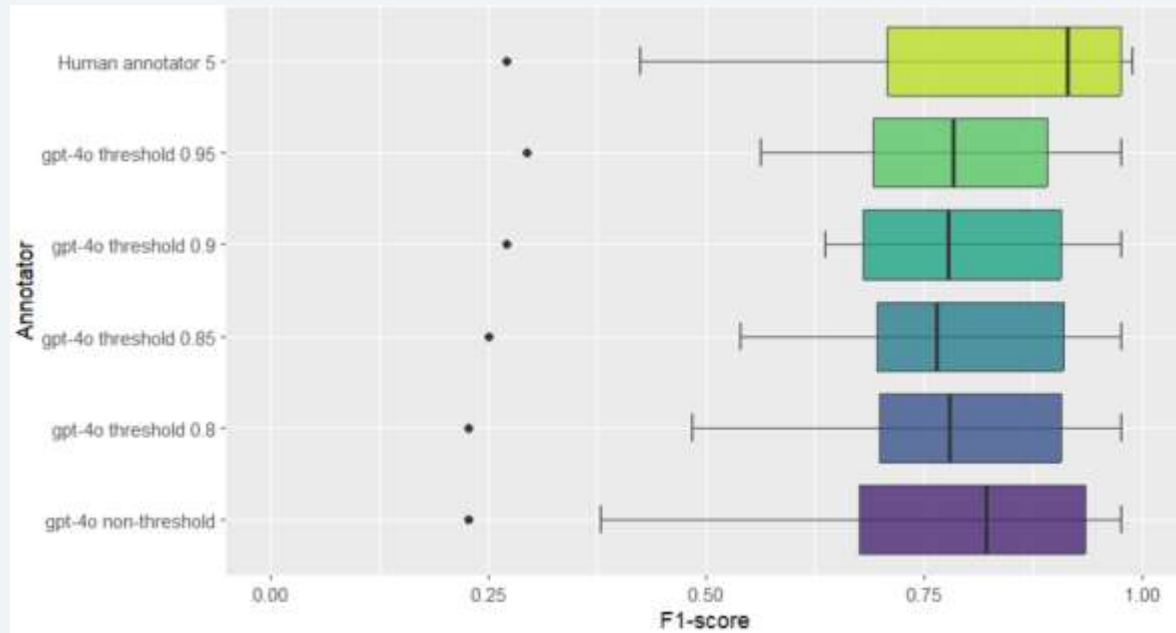
p-values are reported in each box.
Confidence Interval: 95%



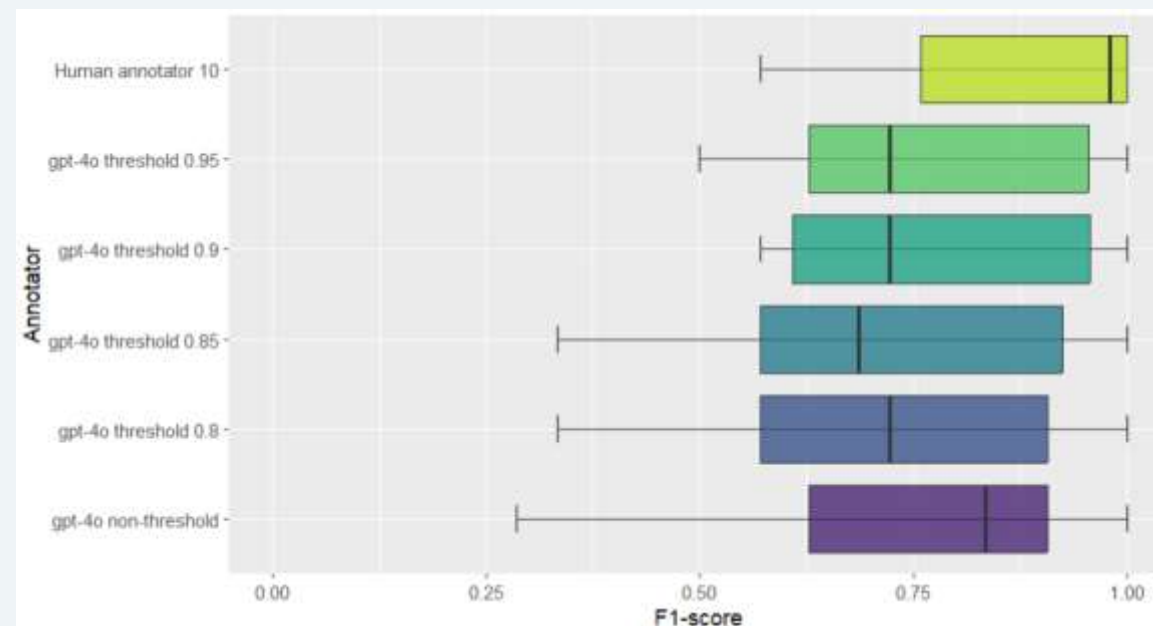
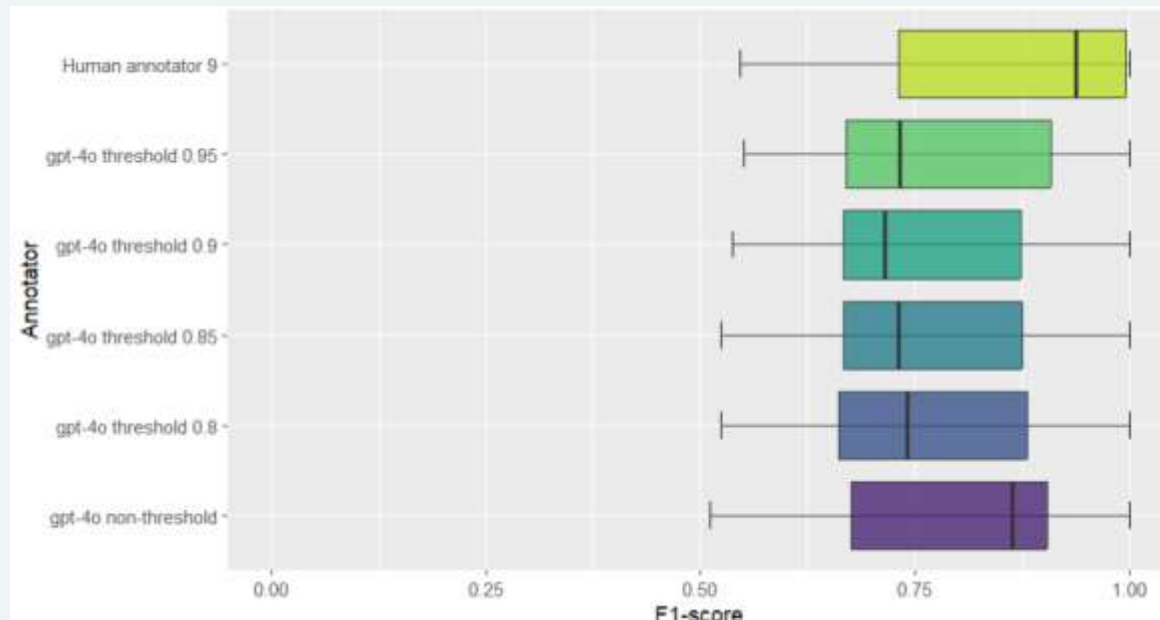
Whole-text level annotations



Whole-text level annotations

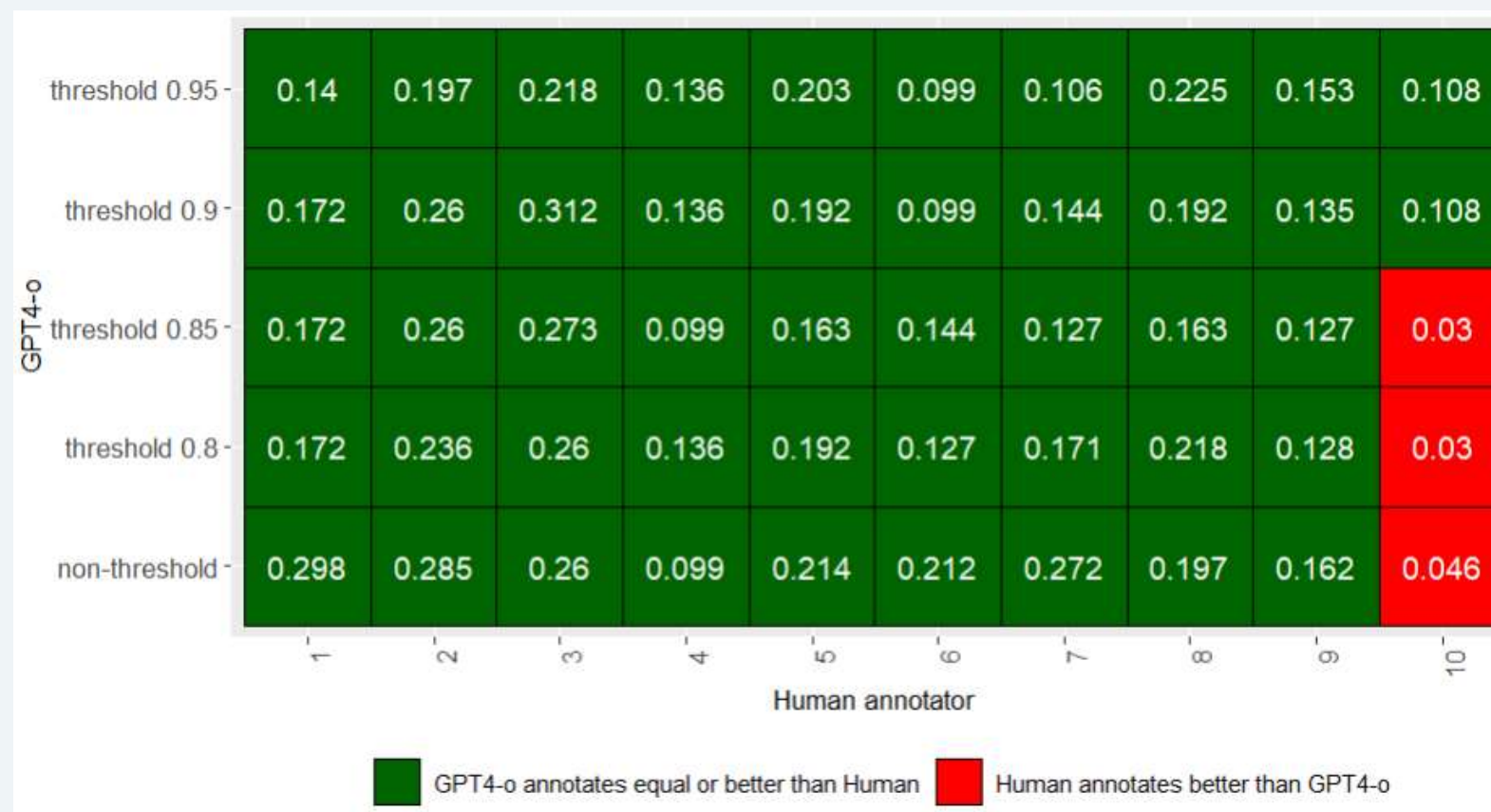


Whole-text level annotations



Whole-text level annotations

p-values are reported in each box.
Confidence Interval: 95%



Conclusion

- Context is important for GPT to make a decision.
- Context is important to establish a log_probability of a chosen category.